Chapter 10

Straight Lines

Only One Option Correct Type Questions

1. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are **[IIT-JEE-2007 (Paper-2)]**

   (A) $\left(\frac{4}{3}, 3\right)$ (B) $\left(3, \frac{2}{3}\right)$

   (C) $\left(3, \frac{4}{3}\right)$ (D) $\left(\frac{4}{3}, \frac{2}{3}\right)$

2. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2\,CD$. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is **[IIT-JEE-2007 (Paper-2)]**

   (A) 3 (B) 2

   (C) $\frac{3}{2}$ (D) 1

3. Let a and b non-zero real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents **[IIT-JEE-2008 (Paper-1)]**

   (A) Four straight lines, when $c = 0$ and a, b are of the same sign

   (B) Two straight lines and a circle, when $a = b$, and c is of sign opposite to that of a

   (C) Two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

   (D) A circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a

4. Consider three points

   $P = (-\sin(\beta - \alpha), -\cos\beta)$, $Q = (\cos(\beta - \alpha), \sin\beta)$

   and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$,

   where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then, **[IIT-JEE-2008 (Paper-2)]**

   (A) P lies on the line segment RQ (B) Q lies on the line segment PR

   (C) R lies on the line segment QP (D) P, Q, R are non-collinear

5. The locus of the orthocentre of the triangle formed by the lines

   $(1 + p)x - py + p(1 + p) = 0$,

   $(1 + q)x - qy + q(1 + q) = 0$,

   and $y = 0$, where $p \neq q$, is **[IIT-JEE-2009 (Paper-2)]**

   (A) A hyperbola (B) A parabola

   (C) An ellipse (D) A straight line

6. A straight line L through the point (3, – 2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is **[IIT-JEE-2011 (Paper-1)]**

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

7. For $a > b > c > 0$, the distance between (1, 1) and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then **[IIT-JEE-2013 (Paper-1)]**

(A) $a + b - c > 0$ (B) $a - b + c < 0$

(C) $a - b + c > 0$ (D) $a + b - c < 0$

8. Consider a triangle Δ whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocentre of Δ is (1, 1), then the equation of the circle passing through the vertices of the triangle Δ is

[JEE (Adv)-2021 (Paper-1)]

(A) $x^2 + y^2 - 3x + y = 0$

(B) $x^2 + y^2 + x + 3y = 0$

(C) $x^2 + y^2 + 2y - 1 = 0$

(D) $x^2 + y^2 + x + y = 0$

Matrix-Match / Matrix Based Type Questions

9. Consider the lines given by **[IIT-JEE-2008 (Paper-2)]**

$L_1 : x + 3y - 5 = 0$

$L_2 : 3x - ky - 1 = 0$

$L_3 : 5x + 2y - 12 = 0$

Match the Statements/Expressions in **Column-I** with the Statements/Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS.

Column-I	Column-II
(A) L_1, L_2, L_3 are concurrent, if	(p) $k = -9$
(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(q) $k = -\frac{6}{5}$
(C) L_1, L_2, L_3 form a triangle, if	(r) $k = \frac{5}{6}$
(D) L_1, L_2, L_3 do not form a triangle, if	(s) $k = 5$

Integer / Numerical Value Type Questions

Question Stem for Question Nos. 10 and 11

Question Stem

Consider the lines L_1 and L_2 defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ, let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S, where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. **[JEE (Adv)-2021 (Paper-1)]**

10. The value of λ^2 is _________.

11. The value of D is _____.

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